「Singular value Decomposition.

Let $V, W$ be inner product space and let $T: V \to W$ be a linear.
Then, there exists orthonormal basis $\{v_1, \cdots, v_n\}$ and $\{u_1, \cdots, u_m\}$ of $V, W$, respectively and positive scalars $\sigma_1 \geq \cdots \geq \sigma_r$ s.t

$$T(v_i) = \begin{cases} \sigma_i u_i & (1 \leq i \leq \operatorname{rank} T) \\ 0 & (i > \operatorname{rank} T) \end{cases}$$

Conversely,

proof. Let $x \in \ker(T^* \circ T)$. $(T^* \circ T)(x) = 0$ implies

$$0 = \langle 0, x \rangle = \langle (T^* \circ T)(x), x \rangle = \langle T(x), T(x) \rangle \Rightarrow T(x) = 0.$$

Therefore, $\ker T^*T \subseteq \ker T$, and the reverse inclusion is clear.
By the Dimension Thm, $\operatorname{rank} T = \operatorname{rank} T^*T$. Moreover, for any non-zero $x \in V$,

$$\langle T^*T(x), x \rangle = \langle T(x), T(x) \rangle = \| T(x) \|^2 \geq 0$$

Thus $T^*T$ is positive semidefinite. Equivalently every eigenvalue of $T^* \cdot T$ is non-negative, and has an orthonormal basis of eigenvectors. put $r = \operatorname{rank} T$. Then,
Denote the positive eigenvalues as

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0$$

and Corresponding eigenvectors as

$$\{v_1, \cdots, v_r, \cdots, v_n\}$$

For each $i$, define $\sigma_i = \sqrt{\lambda_i}$ and $u_i = \frac{1}{\sigma_i} T(v_i)$. Then for any $1 \leq i, j \leq r$,

$$\langle u_i, u_j \rangle = \left\langle \frac{1}{\sigma_i} T(v_i), \frac{1}{\sigma_j} T(v_j) \right\rangle = \frac{1}{\sigma_i \sigma_j} \langle T^*T(v_i), v_j \rangle = \frac{\sigma_i^2}{\sigma_i \sigma_j} \langle v_i, v_j \rangle = \delta_{i,j}.$$

Therefore $\{u_1, \cdots, u_r\}$ is orthonormal. Extending this to $\{u_1, \cdots, u_r, \cdots, u_m\}$, orthonormal basis for $W$.
And $T^*T(v_i) = 0 \Rightarrow T(v_i) = 0$ was proven.

「Singular Value Decomposition.

Let $A \in M_{m\times n}$ with rank $A = r$, with singular values $\sigma_1 \geq \cdots \geq \sigma_r$.
Define a Matrix in $M_{m\times n}$ as

$$\Sigma_{ij} = \begin{cases} \sigma_i & \text{if } i=j<r \\ 0 & \text{otherwise} \end{cases}$$

Then, there exists unitary $U \in M_{m\times m}$ and unitary $V \in M_{n\times n}$ s.t

$$A = U\Sigma V^*$$

$$U = \begin{pmatrix} u_1 & \cdots & u_m \end{pmatrix}, \quad V = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$$

「The Polar Decomposition.

For any $A \in M_{n\times n}$, there exists unitary $W$ and $P \geq 0$ s.t

$$A = WP$$

Furthermore, if $A$ is invertible, this representation is unique.

proof. By the SVD, there exists unitary $U, V$ and diagonal $\Sigma \geq 0$ s.t

$$A = U\Sigma V = \underbrace{UV}_{W}\,\underbrace{V^*\Sigma V}_{P}$$

To show uniqueness, suppose $A$ is invertible and

$$A = W_1P_1 = W_2P_2 \qquad W_1, W_2 \text{ unitary}, \ P_1, P_2 \geq 0$$

Then, $W_2^*W_1 = P_2P_1^{-1}$ unitary,

$$I = (P_2P_1^{-1})^*(P_2P_1^{-1}) = P_1^{-1}P_2^2P_1^{-1} \Rightarrow P_1^2 = P_2^2. \Rightarrow P_1 = P_2.$$